Python for Supply Chain

Session 2 · Wed 18 Nov 2026

Shelves, cards and choices

Today's question

Which order lines are big?

A list holds a row of values

IN DAILY LIFE

A row of bins on one shelf.

IN EXCEL

One column.

THE LIMIT

The first bin is number zero.

```
quantities = [6, 6, 8, 6, 6, 2, 6]
print(quantities[0])
print(quantities[2])
print(quantities[-1])
```

len, sum, min and max measure the list

IN DAILY LIFE

Count the boxes, add them, find the smallest and largest.

IN EXCEL

COUNT, SUM, MIN, MAX.

THE LIMIT

Same idea, new spelling.

```
print(len(quantities))
print(sum(quantities))
print(min(quantities))
print(max(quantities))
```

A dict is a product card with labels

IN DAILY LIFE

A product card: field name to value.

IN EXCEL

One row with headers.

THE LIMIT

Look up by name, not by position.

```
order_line = {
    "quantity": 6,
    "country": "United Kingdom",
}
print(order_line["quantity"])
```

A comparison asks a yes or no question

IN DAILY LIFE

Is this more than that?

IN EXCEL

=A2>10

THE LIMIT

= stores a value. == asks a question.

```
quantity = order_line["quantity"]
is_big = quantity >= 100
print(is_big)
print(quantity == 6)
```

A packing rule becomes if, elif, else

IN DAILY LIFE

A packing rule.

IN EXCEL

=IF(...)

THE LIMIT

The indent is part of the rule.

```
if quantity >= 100:
    print("Big")
elif quantity >= 12:
    print("Medium")
else:
    print("Small")
```

Now in Colab

Notebook	s2_shelves_cards.ipynb
In class	List, len, dict, comparison, if
Before you type	Predict the output first

Three things you can do now

Store a row

a list of quantities

Measure it

len, sum, min and max

Make a choice

if, elif, else

Homework and daily recall

Homework

1. Add rows 8 to 10 from the handout
2. Redo len, sum, min and max
3. Write one if rule about prices

Daily recall

1. Three questions a day
2. Thursday to Sunday